Jason Jaeseung Kim

Biomedical Engineering Undergraduate | Medical AI, Medical Imaging and Deep Learning
xmflak20@gmail.com | importjason.github.io | U.S. Citizen | Seoul, South Korea

RESEARCH FOCUS

Deep learning for medical and healthcare applications, with experience in 3D medical imaging, quantitative analysis, computational MRI, patient-facing AI, and efficient model development. Interested in building reliable systems that turn biomedical data into clinically meaningful information.

EDUCATION

Yonsei University - B.S. Candidate in Biomedical Engineering *Mar 2021 - Expected Feb 2027*

Mirae Campus | Second Major in Electrical and Electronic Engineering, Sinchon Campus

GPA	4.50 Scale	Percentile	Department Rank
4.03 / 4.30	4.18 / 4.50	97.90 / 100	1 / 49

Selected coursework: Image Processing and Applications; Digital Signal Processing; Signals and Systems; Electromagnetics I; Digital Logic Laboratory; Computer Programming; Engineering Mathematics; Deep Learning Laboratory; Public Health and Medical Big Data Analysis.

RESEARCH EXPERIENCE

Undergraduate Researcher, Advanced Image Processing Techniques Lab, Yonsei University *Dec 2023 - Feb 2026*

Advisor: Prof. Joon Yul Choi | Mirae Campus

- Conducted two first-author studies applying deep learning and quantitative analysis to medical-imaging problems; project-specific details are withheld before submission at the advisor's request.

- Developed and evaluated 3D deep-learning workflows for medical-image prediction, quantitative validation, and downstream scientific analysis.

- Built reproducible volumetric imaging pipelines for registration, masking, resampling, normalization, quality control, and downstream analysis.

- Investigated model robustness and data variability across imaging sequences, scanners, vendors, and acquisition conditions, including harmonization-oriented analyses.

- Performed SPM12-based voxel-wise analyses with covariate adjustment and multiple-comparison correction; prepared manuscript text, figures, tables, and research summaries.

Undergraduate Researcher, Bio-Metamaterials Lab, Yonsei University *Jul 2022 - Aug 2022*

Advisor: Prof. Seung Ho Choi | Mirae Campus

- Participated in the initial installation and setup of an ellipsometer and analyzed the compositional and optical properties of biomaterials.

- Built a prototype tensile-strength measurement device and implemented a MATLAB-based user interface for measurement verification.

MANUSCRIPTS AND PRESENTATIONS

- Two first-author journal manuscripts in preparation applying deep learning and quantitative analysis to medical imaging; project titles, methods, results, and figures are withheld before submission.

- Oral Presentation and Outstanding Paper Award, Oral Presentation Session, Korean Society of Medical and Biological Engineering Fall Conference, Nov 2025.

- Poster, NEURO2026, Jul 2026.

SELECTED MEDICAL AI PROJECT AND LEADERSHIP

Project Lead, CareProofRecord - AI-Based Medical Explanation Video Platform *Jun 2025 - Apr 2026*

- Led the planning and development of a medical AI service concept to reduce information-transfer errors and improve patient understanding during surgical and clinical explanations.

- Designed a patient-facing video-generation service architecture intended to mitigate hallucination risk and preserve explanatory context over extended interactions.

- Received KRW 1,000,000 in university startup support after review by the university entrepreneurship program.

- Explored potential clinical use cases with regional medical institutions and other prospective users.

ADDITIONAL PUBLIC TECHNICAL IMPLEMENTATION

Computational MRI Reconstruction and Motion-Correction Reproduction *2026*

Independent computational MRI implementation using public datasets

- Reproduced and analyzed reconstruction and motion-correction pipelines, including image formation, coil-sensitivity estimation, motion simulation, and iterative reconstruction.

- Conducted failure analysis to identify the effects of undersampling, domain shift, calibration limitations, and reconstruction assumptions.

- Documented implementation decisions and reproducibility constraints for subsequent research development.

TECHNICAL SKILLS

Programming	Python; MATLAB; Bash; Linux
Medical AI and Deep Learning	PyTorch; 3D deep learning; image-to-image regression; classification; model explainability; efficient inference; quantitative evaluation
Medical Imaging and Neuroimaging	NIfTI; FSL; ANTs; SPM12; registration; masking; resampling; normalization; VBM; voxel-wise analysis
Statistics and Research Methods	Covariate-adjusted GLM; FWE correction; ROI- and voxel-level analysis; reproducible experimental design

SCHOLARSHIPS

- National Science and Engineering Scholarship, Korea Student Aid Foundation - full tuition (Spring 2023, Fall 2023, Spring 2026).

- Hankyungbum Scholarship, Cheongpa-Hankyungbum Scholarship Foundation, Yonsei University - full tuition (Spring 2022, Fall 2022).

- Academic Excellence Scholarship, Yonsei University (Fall 2021).

HONORS AND AWARDS

Research: Outstanding Paper Award, KOSOMBE Fall Conference, Oral Presentation Session, 2025 **Academic:** Academic Merit Awards, Yonsei University, 2021-2023 **Programming:** Excellence Award, Wise Coding Life Algorithm Competition, Yonsei University, 2022	**Programming:** Grand Prize, ASC Programming Contest, 2021 **Entrepreneurship:** Grand Prize, MEDICI+ Startup Competition, Yonsei University Wonju LINC+ Project Group, 2021 **Entrepreneurship:** Honorable Mention, WAVE-LAB Startup Competition, 2025; Encouragement Award, Post Corona-19 Startup Challenge, 2022